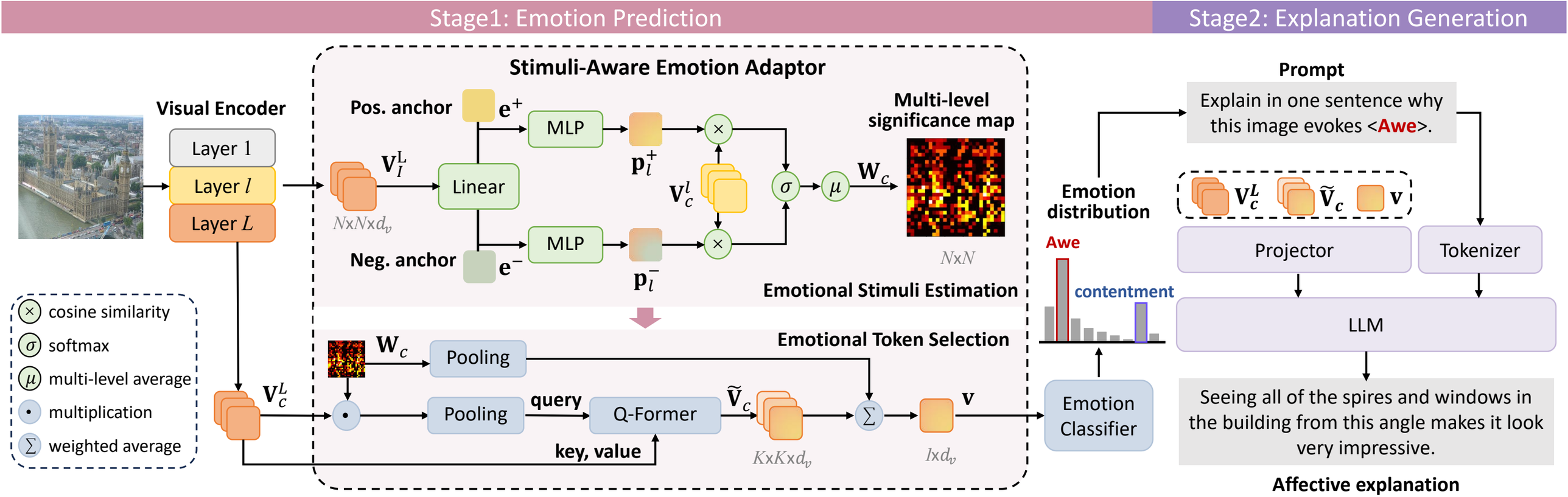
Stage1: Emotion Prediction
Stage2: Explanation Generation
Visual Encoder
Layer 1
Layer l
Layer L
Stimuli-Aware Emotion Adaptor
Pos. anchor
$\mathbf{e}^+$
Neg. anchor
$\mathbf{e}^-$
$\mathbf{V}_I^L$
$NxNxd_v$
Linear
MLP
MLP
$\mathbf{p}_l^+$
$\mathbf{p}_l^-$
$\mathbf{V}_c^l$
σ
μ
$\mathbf{W}_c$
Multi-level significance map
NxN
Emotional Stimuli Estimation
Emotional Token Selection
$\mathbf{W}_c$
Pooling
Pooling
query
Q-Former
key, value
$\tilde{\mathbf{V}}_c$
$KxKxd_v$
$\mathbf{V}_c^L$
$\mathbf{v}$
Ixd_v
cosine similarity
softmax
multi-level average
multiplication
weighted average
Emotion Classifier
Emotion distribution
Awe
contentment
Prompt
Explain in one sentence why this image evokes <Awe>.
$\mathbf{V}_c^L$
$\tilde{\mathbf{V}}_c$
$\mathbf{v}$
Projector
Tokenizer
LLM
Seeing all of the spires and windows in the building from this angle makes it look very impressive.
Affective explanation